

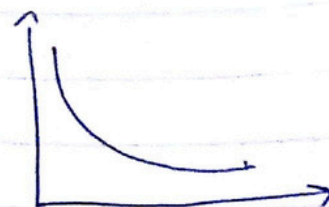
SUBJECT

SECTION

Signal [mid]

- 1 Find the Responses of unit step function $e^{-at} u(t)$ if $t > 0$, $a > 0$. 6 marks

$$e^{-at} u(t) = \begin{cases} e^{-at} & , t \geq 0 \\ 0 & , t < 0 \end{cases}$$



- 2 Define?

1 stable and non stable system. 12 marks

2 CT-LTI system

3

4

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.....
.....

- 3 Find $x(t)$ mathmatically and graph and find energy and power for $x(-3t+2)$. 14 marks

- 4 Find the output Response of $x(n) = [1, -2, 3]$
 $h(n) = [1, 3, 2, 2]$

8 marks



- Consider the signal $x(t) = 4.\cos^2(30\pi t + \pi/8) + 2.\cos(40\pi t - 3\pi/4)$. Is the signal periodic? If so, find the period.
- For the signal shown in Fig. 1, plot:
 - $x(0.5t + 2)$.
 - $2 + 2.x(-2t)$.
 - $2.x(4 - 2t)$.
 - The even and odd parts of $x(t)$.

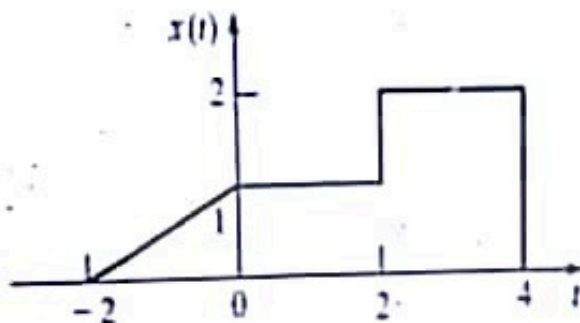


Figure 1

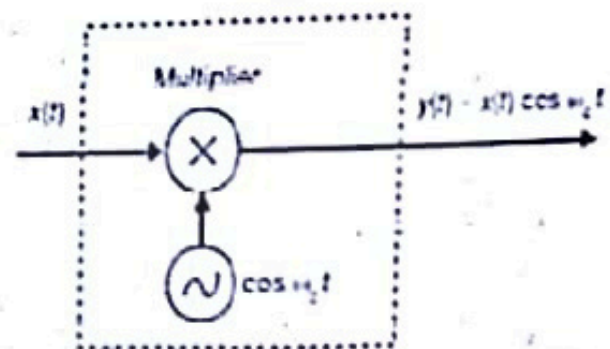


Figure 2

- Consider a system shown in Fig. 2. Determine whether it is: (a) Memoryless, (b) causal, (c) linear, (d) fixed, or (e) stable. Why?
- Sketch the following signals:
 - $x_1(t) = r(t + 4) - 2.r(t) + r(t - 4) + 2.\text{rect}(0.5t)$.
 - $x_2(t) = 2.u(t + 3)u(3 - t) + (1 - t^2).u(t + 1).u(1 - t)$.
- Consider $h(t) = 2.\text{rect}((t - 2)/4)$ and $x(t) = 2.r(t - 2)$.
 - Plot $h(t)$ and $x(t)$.
 - Determine $y(t) = h(t) * x(t)$.
 - Is the system BIBO stable? Why?
- Evaluate the following integrals:

a. $\int_{-\infty}^{\infty} \cos(2 - t).\delta(t - 2)dt.$

b. $\int_{-\infty}^0 (t - 2)^2.\delta(t + 0.5)dt.$

c. $\int_0^{20} (4 + \exp(0.1t)).\delta(2t - 30)dt.$

P.1 $x(t) = 4\cos^2(30\pi t + \pi/8) + 2\cos(40\pi t - 3\pi/4)$
 $= 2 + 2\cos(60\pi t + \pi/8) + 2\cos(40\pi t - 3\pi/4)$

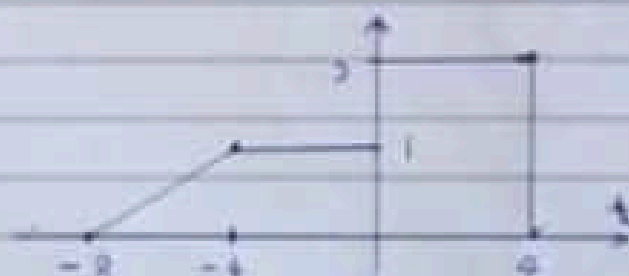
$$T_{01} = \frac{2\pi}{60\pi} = \frac{1}{30} \text{ sec} \rightarrow T_{02} = \frac{2\pi}{40\pi} = \frac{1}{20} \text{ sec}$$

$$\frac{T_{02}}{T_{01}} = \frac{1/20}{1/30} = \frac{3}{2} \rightarrow x(t) \text{ is periodic signal.}$$

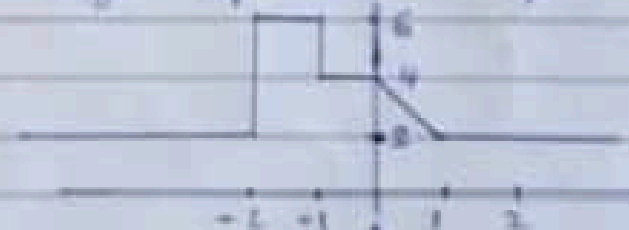
$$T_0 = 2T_{02} = 3T_{01} = \frac{1}{10} = 0.1 \text{ sec.}$$

P.2 $x(t) = \frac{1}{2}r(t+2) - \frac{1}{2}r(t) + u(t-2) - 2u(t-4)$
 $= 0.5r(t+2) - \frac{1}{2}r(t) + u(t-2) - 2u(t-4)$
0.5

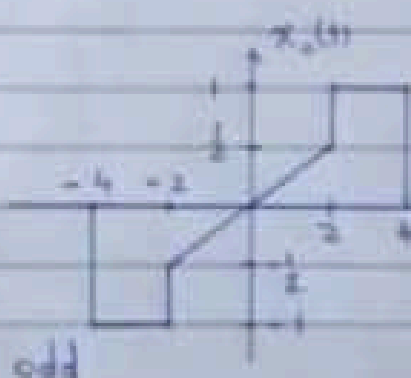
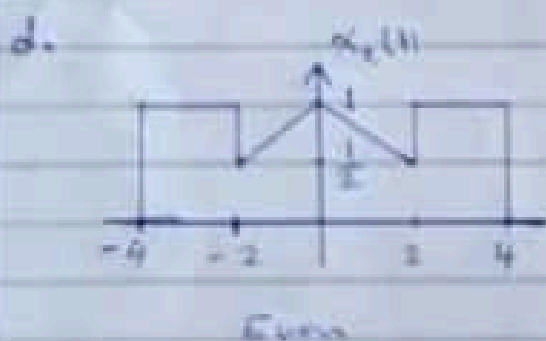
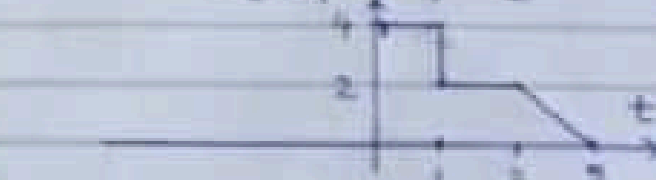
a. $x(0.5t+2) =$
 $x(0.5(t+4))$
 0.5: expansion
 + 4: shift left by 4



b. $2 + 2 \cdot x(-2t)$
 compression
 amplitude scaling
 shift



c. $2x(4-2t) = 2x(1-2(t-2))$



P5 c. $\int_{-\infty}^{\infty} |h(t)| dt = \int_0^4 2 dt = 2(4-0) = 8 < \infty$

The system is BIBO stable.

P6 a. $\int_{-\infty}^{\infty} \cos(2-t) \delta(t-2) dt = \cos(2-2) = \cos(0) = 1$

b. $\int_{-\infty}^{\infty} (t-2)^2 \delta(t+0.5) dt = (-0.5-2)^2 = (-2.5)^2 = 6.25$

c. $\int_0^{20} (4 + e^{0.1t}) \delta(2t-30) dt = \int_0^{20} (4 + e^{0.1t}) \delta(2(t-15)) dt = \frac{1}{2} (4 + e^{1.5}) = 4.24$

P3 $y(t) = x(t) \cos(\omega_0 t)$

a. Memoryless. $y(t)$ It depends on present time of input only.

b. Causal $y(t)$ does not depend on future time of $x(t)$.

c. Linear. $y(t) = T[a_1 x_1(t) + a_2 x_2(t)]$
 $= a_1 x_1(t) \cos(\omega_0 t) + a_2 x_2(t) \cos(\omega_0 t)$
 $= a_1 y_1(t) + a_2 y_2(t)$

d. Not fixed (time variant).

$$\begin{aligned} y(t-t_0) &= x(t-t_0) \cos(\omega_0 (t-t_0)) \\ &= x(t-t_0) [\cos(\omega_0 t) \cos(\omega_0 t_0) + \sin(\omega_0 t) \sin(\omega_0 t_0)] \\ &\neq x(t-t_0) \cos(\omega_0 t) \end{aligned}$$

e. Stable:

$$|y(t)| = |x(t) \cos(\omega_0 t)| < |x(t)|$$

If $x(t)$ is bounded then $y(t)$ is bounded.

→ The system is BIBO stable.

P2 b. $\int_{-\infty}^{\infty} |h(t)| dt = \int_0^4 2 dt = 2(4-0) = 8 < \infty$

The system is BIBO stable.

P2 a. $\int_{-\infty}^{\infty} \cos(t-2) \delta(t-2) dt = \cos(2-2) = \cos(0) = 1$

b. $\int_{-\infty}^{\infty} (t-2)^3 \delta(t+0.5) dt = (-0.5-2)^3 = (-2.5)^3 = -6.25$

c. $\int_0^{\infty} (4 + e^{15t}) \delta(2t-30) dt = \int_0^{\infty} (4 + e^{15t}) \delta(t-15) dt = \frac{1}{2} (4 + e^{15 \cdot 15}) = 9.24$

P3 $y(t) = x(t) \cos(\omega t)$

a. Memoryless. $y(t)$ depends on present time of input only.

b. Causal. $y(t)$ does not depend on future time of $x(t)$.

c. Linear.
$$y(t) = T[a_1 x_1(t) + a_2 x_2(t)]$$

$$= a_1 x_1(t) \cos(\omega t) + a_2 x_2(t) \cos(\omega t)$$

$$= a_1 y_1(t) + a_2 y_2(t)$$

d. Not fixed (time variant)

$$y(t-t_0) = x(t-t_0) \cos(\omega(t-t_0))$$

$$= x(t-t_0) [\cos(\omega t) \cos(\omega t_0) + \sin(\omega t) \sin(\omega t_0)]$$

$$\neq x(t-t_0) \cos(\omega t)$$

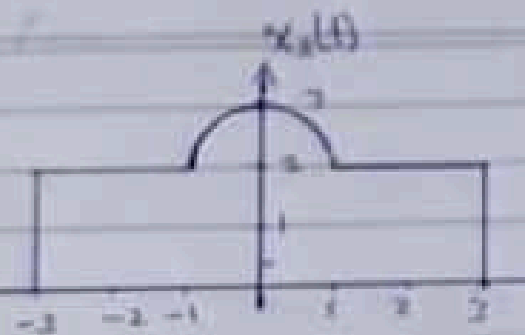
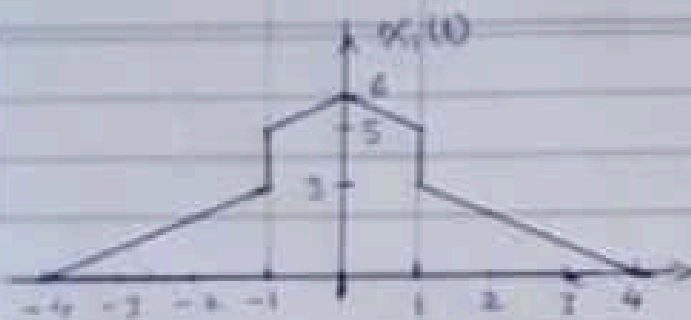
e. Stable:

$$|y(t)| = |x(t) \cos(\omega t)| \leq |x(t)|$$

If $x(t)$ is bounded then $y(t)$ is bounded.

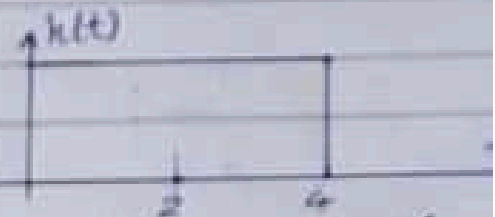
→ The system is BIBO stable.

P.4



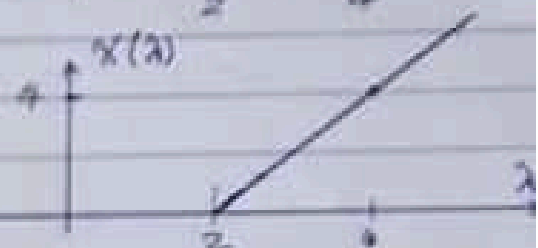
P.5 a. $h(t) = 2 \text{rect}(\frac{t-2}{4})$

$$x(t) = 2r(t-2) = \begin{cases} 2t-4 & t > 2 \\ 0 & t < 2 \end{cases}$$



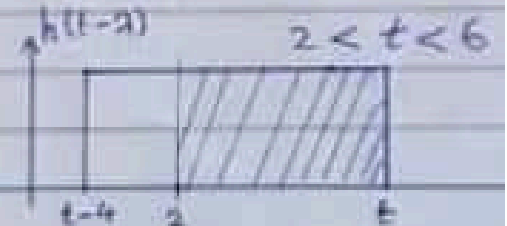
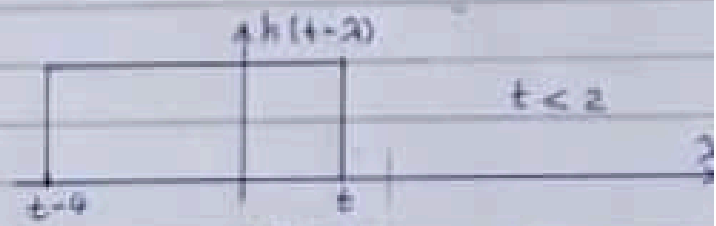
b. $y(t) = \int_{-\infty}^{\infty} x(\lambda) h(t-\lambda) d\lambda$

* $t < 2$:
 $y(t) = 0$



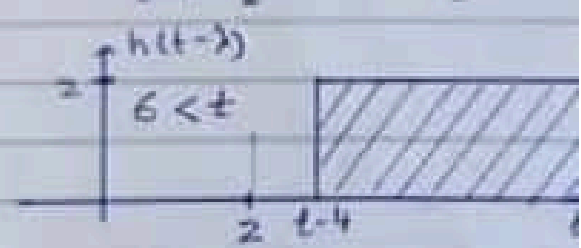
* $2 < t < 6$

$$\begin{aligned} y(t) &= \int_{t-4}^t (4\lambda - 8) d\lambda \\ &= [2\lambda^2 - 8\lambda]_{t-4}^t \\ &= 2t^2 - 8t + 8 \end{aligned}$$



* $6 < t$

$$\begin{aligned} y(t) &= \int_{t-4}^t (4\lambda - 8) d\lambda \\ &= [2\lambda^2 - 8\lambda]_{t-4}^t \\ &= 16t - 64 = 16(t-4) \end{aligned}$$



$$y(t) = \begin{cases} 0 & t < 2 \\ 2t^2 - 8t + 8 & 2 < t < 6 \\ 16(t-4) & 6 < t \end{cases}$$